AI Researcher | LLM Systems | Inference Optimization | NLP | Agentic AI

HIGHLIGHTS

- Total 10 YoE in AI solutions development comprising 3 YoE in leading Data Science teams
- Top Talent in Reliance Jio R&D for two consecutive years with an A+ rating (2022-2024)
- Former co-founder and AI R&D Head of Sports Injury Management Startup (Dallas, US)
- NLP (Sentiment Information Mining) Data Science Hackathon Finalist - Runner-up
- Oral paper accepted at EuroMLSys 2026 (co-located with EuroSys, Edinburgh) for PROTEUS: Lagrangian RL-based SLA-aware LLM Router

SKILLS

Research Areas : LLM inference optimization, SLA-aware routing, Lagrangian RL, constrained MDPs, KV cache architectures, speculative decoding, transformer internals (MHA/GQA/MQA/MLA), MoE routing, RLHF

GenAI & LLMs : DeepSeek R1/V3, Qwen, LLaMA 3.x, GPT-4o, Claude 3.x, Mistral, BERT, T5, RAG, Vector Databases, LoRA, QLoRA, PEFT, Weaviate, FAISS

Inference & Serving : vLLM, TensorRT-LLM, Triton Inference Server, TorchServe, ONNX, continuous batching, paged KV cache, prefill-decode disaggregation, FP16/BF16/INT8, mixed precision, speculative decoding

Quantization & Compression : GPTQ, AWQ, zero-calibration quantization, pruning, knowledge distillation, SFT, instruction tuning

Agentic Systems : LangGraph, LangChain, LlamaIndex, AutoGen, CrewAI, DSPy, Claude Code, OpenCLAW, MCP, A2A, ACP, tool/function calling, multi-agent orchestration

ML Frameworks : PyTorch, TensorFlow, HuggingFace Transformers, Scikit-Learn, NLTK, spaCy, Gensim, FastText, PySpark, CUDA, OpenCV

Deployment & MLOps : Docker, Kubernetes, KubeFlow, MLFlow, FastAPI, Flask, Prometheus, Logstash

Evaluation : MT-Bench, MMLU, custom SLA/cost Pareto frontier analysis, latency profiling, throughput characterization, SHAP/XAI

RESEARCH

- **PROTEUS:** SLA-Aware LLM Router using Lagrangian dual-variable RL for simultaneous latency and cost SLA enforcement across heterogeneous model tiers. Oral presentation accepted at EuroMLSys 2026 (co-located with EuroSys, Edinburgh).
- **KV-Hub:** Cross-model KV cache transfer architecture via depth-normalized terminal layer alignment, reducing redundant prefill computation in multi-model serving pipelines. Patent filed.
- **Towards Open Domain Automatic Information Extraction in Hindi (AAAI Workshop):** Hybrid LLaMA-2-70B + CRF decoder system benchmarked against StanfordCoreNLP, LLaMA2-70B, and GPT-4o; achieved 2.5% gain above baseline on Hindi NER/IE.
- **MS Thesis (LJMU):** Neuroticism Alignment Detection for Misogyny Aggression via Hierarchical Explanation Method (HEDGE + SHAP-based XAI applied to toxic-speech classification).

EMPLOYMENT HISTORY

AI Research Lead - LLM Efficiency and Optimisation, Quantiphi R&D Bengaluru *Feb 2025 - Present*

- Designed and published PROTEUS, a Lagrangian RL-based LLM routing framework enforcing joint latency and cost SLAs dynamically; accepted for oral presentation at EuroMLSys 2026.
- Architected KV-Hub, a cross-model KV cache reuse system exploiting depth-normalized terminal layer alignment to reduce redundant prefill computation in multi-model serving clusters.

- Developed and deployed production agentic routing pipeline achieving 2.5x inference cost reduction through SLM selection and intelligent tool orchestration for customer-support and finance assistants.
- Standardized multi-agent orchestration patterns (tool calling, retrieval, human-in-the-loop approval) for production agentic workflows.
- Filed two novel IP patents (KV-Hub architecture and PROTEUS dual-variable SLA controller) with comprehensive prior art analysis.

Senior AI Scientist, Reliance Jio Hyderabad *Dec 2021 - Jan 2025*

- Jio Telecommunication Query Resolution System - Led an NLP team of six AI engineers to develop an in-house proprietary LLM-based query resolution engine in Indic Languages. LLaMA 3.1-8B was instruction fine-tuned using SFT + LoRA with an in-house dataset of Hindi, Marathi and Telugu. Guided optimization (Pruning + Quantization) and hosting on vLLM inference server, reducing inference cost by 12x.
- Research Work - Towards Open Domain Automatic Information Extraction in Hindi (AAAI Workshop) - Developed novel hybrid Information Extraction system (Part of Speech tagging, Sense Disambiguation) with LLaMA2-70B + CRF decoder benchmarked against StanfordCoreNLP, LLaMA2-70B, and OpenAI API-4o. Achieved 2.5% gain above baseline metrics.

Data Scientist, Legato Health Technologies - Anthem Inc Hyderabad *Sept 2020 - Nov 2021*

- Led development of an end-to-end meeting transcription system using NVIDIA NeMo toolkit, implementing multi-speaker diarization with TitaNet embeddings and MSDD (Multi-Scale Diarization Decoder), achieving 95% speaker identification accuracy for medical conversations.
- Architected comprehensive NLP pipeline processing diarized text through fine-tuned transformers (BERT for sentiment/NER, T5 for summarization), automating medical Minutes of Meeting generation and reducing documentation time by 70%.

Deep Learning Software Engineer, Inspirit-IoT Technologies Chennai *Jan 2019 - Aug 2020*

- Collaborated with UIUC PhD researchers to develop DNNOptimizer, a proprietary neural network compression framework that surpassed TensorRT benchmarks for vision models (ResNet-50, YOLOv3, FaceNet) on Intel Arria FPGAs.
- Implemented and optimized object detection and facial recognition pipelines through hardware-aware compression techniques for FPGA-specific optimizations.

Machine Learning Engineer, Virtusa Consulting Hyderabad *Nov 2016 - Dec 2018*

- Pioneered deep learning research for insurance claims classification using hybrid LSTM/GRU models with custom text preprocessing, achieving 87% accuracy; designed insurance-specific embeddings and hierarchical classification improving model interpretability by 40% and reducing processing errors by 25%.
- Architected production ETL pipeline handling 100K+ daily claims, contributing to a classification patent and implementing domain-specific NLP models; integrated real-time fraud detection achieving 92% accuracy; designed scalable microservices architecture reducing processing latency by 60% while maintaining 95% uptime.

PATENTS

- Triad Three Coordinate based degree of turns and frequency of turns computation for kinematics sports movement - Patent Number (India) - 202311036108
- Kinematics Spatial Vision Grading System for Prediction of Anterior Cruciate Tears - Patent Number (India) - 202311036107

EDUCATION

Oct 2020 - Jul 2022 **MS by Research (NLP) - Data Science,** Liverpool John Moores University

Thesis: Neuroticism Alignment Detection for Misogyny Aggression via Hierarchical Explanation Method. Analyzed the HEDGE algorithm's explanation weights for alignment with neuroticism, a key trait in Incel behavior.

Jul 2012 - Jul 2016 **B.Tech - Robotics and Control Engineering,** NIT Jalandhar

PUBLICATIONS & TALKS

- PROTEUS: Lagrangian RL-based SLA-Aware LLM Router - Oral presentation, EuroMLSys 2026, Edinburgh (April 2026)

- Towards Open Domain Automatic Information Extraction in Hindi - AAAI Workshop
- Technical blog: bayesiansapien.tech (Substack + LinkedIn) covering LLM inference optimization, transformer internals, and agentic AI systems